Ex No: 1d

IMPLEMENTATION OF A* SEARCH ALGORITHM

Aim:

To implement A* Search Algorithm

Procedure:

Program:

```
import heapq

class Node:
    def __init__(self, state, parent, g, h):
        self.state = state
        self.parent = parent
        self.g = g  # Cost to reach the current node
        self.h = h  # Heuristic cost from current node to goal
        self.f = g + h  # Total cost (f = g + h)

    def __lt__(self, other):
        return self.f < other.f


def a_star_search(start, goal, heuristic, neighbors):
    open_list = []
```

```
    closed_list = set()

    start_node = Node(start, None, 0, heuristic(start, goal))
    heapq.heappush(open_list, start_node)

    while open_list:
        current_node = heapq.heappop(open_list)

        if current_node.state == goal:
            # Reconstruct the path from start to goal
            path = []
            while current_node:
                path.append(current_node.state)
                current_node = current_node.parent
            return path[::-1]

        closed_list.add(current_node.state)

        for neighbor, cost in neighbors(current_node.state):
            if neighbor in closed_list:
                continue

            g = current_node.g + cost
            h = heuristic(neighbor, goal)
            neighbor_node = Node(neighbor, current_node, g, h)

            if not any(open_node.state == neighbor and open_node.f <= neighbor_node.f for
open_node in open_list):
                heapq.heappush(open_list, neighbor_node)

    return None  # If no path is found
```

```
# Example of usage

def heuristic(state, goal):
    # Simple Manhattan distance heuristic for a grid (assuming coordinates)
    return abs(state[0] - goal[0]) + abs(state[1] - goal[1])

def neighbors(state):
    # Define neighbors as adjacent grid cells (up, down, left, right)
    x, y = state
    return [
        ((x + 1, y), 1),  # Right
        ((x - 1, y), 1),  # Left
        ((x, y + 1), 1),  # Up
        ((x, y - 1), 1)   # Down
    ]



start = (0, 0)
goal = (3, 3)

path = a_star_search(start, goal, heuristic, neighbors)
print("Path from start to goal:", path)
```

Output:

For `start = (0, 0)` and `goal = (3, 3)`:
Path from start to goal: [(0, 0), (1, 0), (2, 0), (3, 0), (3, 1), (3, 2), (3, 3)]